ELEVEN AND TWELVE LONELY RUNNERS

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Abstract. The Lonely Runner Conjecture asserts that for any number of runners moving at distinct constant speeds around a unit circle from a common starting point, each runner is lonely at some time. A runner is said to be lonely if its distance from every other runner exceeds $1/(k+1)$, where $k+1$ is the total number of runners. A computational method derived by Rosenfeld [4] and Trakulthongchai [7] verified the conjecture for up to ten runners In this work, we provide computer-assisted proofs that the Lonely Runner Conjecture holds in the 11 and 12 runners cases.

Introduction

The Lonely Runner Conjecture, attributed to Wills [9], states that for any $n > 2$, if n runners are running with constant distinct speed around a unit circle, each runner would be eventually *lonely*. Formally that means for each runner i, there is a point in time t such that no runner is within distance $\frac{1}{n}$ of runner i. By using the symmetry of the problem, one may fix a runner to be stationary and measure all other speeds relative to it. This yields an equivalent formulation as presented in Conjecture 1.

Conjecture 1. *Let $v_1, \ldots, v_k$ be positive real numbers. Then there exists a real number t such that*

$$\|tv_i\| \geq \frac{1}{k+1} \quad \textit{for all } i = 1, \ldots, n,$$

The conjecture is trivial for 2 and 3 runners. However, as the number of runners increases, the complexity grows exponentially. While showing the proof for the case of six runners, Bohman, Holzman, and Kleitman [1] showed that it is enough to considered integer speeds, reducing the problem from a continuous to a discrete one. Moreover, Tao [6] showed that it is possible to computationally verify the conjecture by only considering integer speeds up to $n^{O(n^2)}$, turning an infinite problem into a verifiable one. Later on, Malikiosis, Santos, and Schymura [2] reduced the search space drastically by showing that it is enough to consider integer speeds up to only $\binom{n+1}{2}^{n-1}$.

Building on the result of Malikiosis [2], Rosenfeld [4] proved the Lonely Runner Conjecture for 8 runners and Trakulthongchai [7] improved the computational search to assert for 9 and 10 runners.

In this work, we improved the computational method used by Rosenfeld and Trakulthongchai to conclude that the conjecture holds for 11 and 12 runners.

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Rosenfeld's and Trakulthongchai's Computational Method

By Corollary 3 of [4], derived from the result of Malikiosis, Santos, Schymura [2], if the $k-1$ Lonely Runner Conjecture holds, the k lonely runners can be proven by verifying that there are no integral speed set $\{v_1, \ldots, v_k\}$ that is a counterexample to the conjecture for

$$v_1 \ldots v_k < B_k \quad \text{where} \quad B_k := \left(\frac{\binom{k+1}{2}^{k-1}}{k} \right)^k$$

Rosenfeld introduced the idea of attacking the bound by generating multiple small proofs, each of which is a proof that if $v_1, \ldots, v_k$ is a counterexample, then a particular prime p divides $v_1 \cdots v_k$. By accumulating enough proofs for a set of primes P, it follows that

$$\prod_{p \in P} p \;\Big|\; v_1 \cdots v_k.$$

which shows $v_1 \cdots v_k \geq \prod_{p \in P} p > B_k$

Following Rosenfeld's approach, Trakulthongchai [7] defined proper and improper solution sets. The following definition 2 is a reformulation of Definition 3.1 from [7].

Definition 2. *Let p be a prime greater than $k+1$ and $\gcd(p, l) = 1$. Let $\pi : \mathbb{Z}/pl\mathbb{Z} \to \mathbb{Z}/p\mathbb{Z}$ be the natural projection. A speed set $v := \{v_1, \ldots, v_k\}$ is such that $v_i \in \pi^{-1}((\mathbb{Z}/lp\mathbb{Z})^\times)$ for all i. A speed set v is improper modulo lp, write $v \in I(k, l, p)$ if both conditions hold:*

(1) for any $t \in \mathbb{Z}/lp\mathbb{Z}$ there exists i such that $\left\| \frac{tv_i}{lp} \right\| < \frac{1}{k+1}$.
(2) for all i, $\gcd(lp, v_1, \ldots, v_{i-1}, v_{i+1}, \ldots, v_k) = 1$.

and $I(k, l, p)$ denotes the set of all improper speed sets for fixed k, l, p.

Intuitively, the condition states that for every time parameter t modulo p, there is a runner who fails to achieve the required separation distance $\frac{1}{k+1}$ from the stationary runner. Thus the set is a potential obstruction to the Lonely Runner Conjecture when considered modulo p.

He showed in Lemma 3.3 of [7] that if the set of improper speed sets $I(k, l, p)$ is empty for some value of l, then $p \mid v_1 \cdots v_k$ in any counterexample.

Trakulthongchai then proposed an efficient method that allows *lifting* a candidate solution set modulo lp into a candidate solution set modulo clp. Allowing computation to use prime factors of $k+1$ to help speed up the process. For example, with $k = 8$, the idea allows splitting the computation into 3 steps, computing $I(k, 1, p)$ then $I(k, 3, p)$ and then $I(k, 9, p)$.

Main Idea: Quotient modulo p

The framework introduced in [7] has a bottleneck in the first phase of computation, the calculation of $I(k, 1, p)$. Therefore we aimed to exploit the $(\mathbb{Z}/p\mathbb{Z})^\times$ symmetry of $I(k, l, p)$ to reduce the amount of computation needed for $I(k, 1, p)$. Notice that for $l = 1$ and a prime modulus p, the objects of interest are speed sets whose elements lie in the multiplicative group $(\mathbb{Z}/p\mathbb{Z})^\times$.

In order to facilitate reasoning about speed set, let $(v_1, \ldots, v_k)$ be a canonical representation of speed set $\{v_1, \ldots, v_k\}$ such that two set are equal if and only if the canonical representations equal. The representation can be chosen arbitrarily. In this work and the algorithm written in [5], we use $(v_1, \ldots, v_k)$ such that $v_1 < \cdots < v_k$.

Lemma 3. *For any improper speed set $(v_1, \dots, v_k)$ in $I(k, 1, p)$, there exists an improper set $(1, u_2, \dots, u_k)$ in $I(k, 1, p)$ and a number $u \in (\mathbb{Z}/p\mathbb{Z})^\times$ such that $(u, uu_2, \dots, uu_k) = (v_1, \dots, v_k)$.*

Proof. Since v_1 is a speed in a speed set modulo p, $v_1 \in (\mathbb{Z}/p\mathbb{Z})^\times$. Let $u = v_1$ and $u_i = v_1^{-1} v_i$ for each i. Then, $uu_i = v_1 \cdot v_1^{-1} v_i = v_i$. Hence

$$(u, uu_2, \dots, uu_k) = (v_1, v_2, \dots, v_k).$$

It remains to verify that $\{1, u_2, \dots, u_k\}$ is also an improper set. Since $l = 1$, $\gcd(p, v_i) = 1$ for any i, so the gcd condition holds trivially. Therefore, it is enough to show the first condition. To do that, pick any $t \in \mathbb{Z}/p\mathbb{Z}$ and set $t' = tv_1^{-1}$. Since $(v_1, \dots, v_k)$ is improper, there exists some index i such that

$$\frac{1}{k+1} > \left\| \frac{t' v_i}{p} \right\| = \left\| \frac{t v_1^{-1} v_i}{p} \right\| = \left\| \frac{t u_i}{p} \right\|$$

where in particular, $u_1 = 1$. As both conditions in Definition 2 hold for the set $\{1, u_1, \dots, u_k\}$, the set is improper. □

The lemma shows an equivalent relation of improper sets in $I(k, 1, p)$ under multiplication by units. Consequently, when enumerating possible counterexamples modulo p it suffices to consider only *normalized* sets of the form $\{1, u_2, \dots, u_k\}$. This reduces the search space by a factor of p.

When lifting $I(k, l, p)$ to $I(k, cl, p)$ by a multiplier c, the *shadow*, as described in [7], of $I(k, l, p)$ at level cl is the preimage of the coordinate-wise extension of the natural projection $\pi : \mathbb{Z}/clp\mathbb{Z} \to \mathbb{Z}/lp\mathbb{Z}$. Formally

$$\delta_{cl}(S) = (\pi^k)^{-1}(S) \text{ where } \pi^k(v_1, \dots, v_k) := (\pi(v_1), \dots, \pi(v_k))$$

This motivates the following lemma 4 as an extension of lemma 3 for $l > 1$.

Lemma 4. *Let $\gcd(l, p) = 1$ and $\pi : \mathbb{Z}/pl\mathbb{Z} \to \mathbb{Z}/p\mathbb{Z}$ be the natural projection. For any improper set $(v_1, \dots, v_k)$ in $I(k, l, p)$, there is an improper set $(u_1, u_2, \dots, u_k)$ in $I(k, l, p)$ and a number $u \in (\mathbb{Z}/pl\mathbb{Z})^\times$ such that $\pi(u_1) = 1$, i.e. $u_1 \equiv 1 \pmod p$, and*

$$(uu_1, uu_2, \dots, uu_k) = (v_1, \dots, v_k) \text{ as sets in } \mathbb{Z}/lp\mathbb{Z}$$

Proof. Since $\gcd(p, l) = 1$, the restriction $\pi|_{(\mathbb{Z}/pl\mathbb{Z})^\times} : (\mathbb{Z}/pl\mathbb{Z})^\times \to (\mathbb{Z}/p\mathbb{Z})^\times$ is surjective by the Chinese Remainder Theorem. Since $\{v_1, \dots, v_k\}$ is an improper set, $\pi(v_1) \neq 0$, so $\pi(v_1) \in (\mathbb{Z}/p\mathbb{Z})^\times$. By surjectivity, there exists $u \in (\mathbb{Z}/pl\mathbb{Z})^\times$ with $\pi(u) = \pi(v_1)$, and pick $u_i := u^{-1} v_i$ for each i.

This makes $uu_i = v_i$ for all i, so $\{uu_1, \dots, uu_k\} = \{v_1, \dots, v_k\}$. Moreover,

$$\pi(u_1) = \pi(u)^{-1} \pi(v_1) = \pi(v_1)^{-1} \pi(v_1) = 1.$$

It remains to show that $\{u_1, \dots, u_k\}$ is an improper set. For the gcd condition, since $u \in (\mathbb{Z}/pl\mathbb{Z})^\times$ we have $\gcd(u^{-1}, lp) = 1$, so multiplication by u^{-1} preserves all divisibility relations with respect to lp. Hence for each i,

$$\gcd(u_1, \dots, \widehat{u}_i, \dots, u_k, lp) = \gcd(v_1, \dots, \widehat{v}_i, \dots, v_k, lp) = 1,$$

where the last equality holds since $\{v_1, \dots, v_k\}$ is an improper set.

For the second condition, pick any $t \in \mathbb{Z}/pl\mathbb{Z}$ and set $t' = tu^{-1}$ resembling lemma 3. Since $\{v_1, \ldots, v_k\}$ is improper, there exists some index i such that

$$\frac{1}{k+1} > \left\| \frac{t'v_i}{pl} \right\| = \left\| \frac{tu^{-1}v_i}{pl} \right\| = \left\| \frac{tu_i}{pl} \right\|$$

which fulfills condition for $\{u_1, \ldots, u_k\}$ to be improper. □

The lemma shows that every improper set in $I(k, l, p)$ is a unit multiple of an improper set whose first element projects to $1 \in \mathbb{Z}/p\mathbb{Z}$. In particular, it suffices to enumerate improper sets only among those with $\pi(u_1) = 1$ because

$$I(k, l, p) = \emptyset \iff \{\{u_1, \ldots, u_k\} \in I(k, l, p) : \pi(u_1) = 1\} = \emptyset.$$

This reduces the amount of computation needed by roughly a factor of p as it is enough to keep track of $I(k, l, p)/(\mathbb{Z}/p\mathbb{Z})^\times$.

In addition to this normalization, I improved the computational approach of Trakulthongchai by refactoring the implementation to support parallel execution and by introducing more memory-efficient data structures. The new implementation remains faithful to the original algorithmic ideas but is significantly faster in practice.

Introducing intermediate sieves

This idea was developed from the suggestion introduced by Trakulthongchai in section 7 of [7]. In this section, we write $I'(k, l, p) := I(k, l, p)/(\mathbb{Z}/p\mathbb{Z})^\times$.

The cost of lifting $I'(k, l, p)$ to $I'(k, cl, p)$ is proportional to $|I'(k, l, p)| \cdot c^k$, since each improper set modulo lp has c^k candidate preimages modulo lcp to check. In particular, a lift with $c = 2$ costs $|I'(k, l, p)| \cdot 2^k$, whereas a lift with $c = 3$ costs $|I'(k, l, p)| \cdot 3^k$, making the $c = 2$ step significantly cheaper for large k.

Moreover, I observed that lifting with $c = 2$ can reduce the size of the improper set, i.e. $|I'(k, 2l, p)| < |I'(k, l, p)|$ in practice. This observation suggests that using a $c = 2$ lift can minimize $|I'(k, l, p)|$ before passing through the more expensive $c = 3$ lift. Performing an extra cheaper step of $c = 2$ lift can shorten the overall computation time by reducing the work needed to be done in larger mod.

To put numbers to the observation, I have devised two runs to prove the case of $k = 8$. The first one follows Trakulthongchai's method described in Algorithm 4.1 of [7]. The second one involves trying to lift with $c = 2$ twice and keep the smallest set. Specifically, the second method can be described as

(1) Generate $S_1 := I'(8, 1, p)$
(2) Compute $S_2 := I'(8, 2, p)$ by lifting S_1.
(3) If $|S_2| < |S_1|$, compute $S_4 := I'(8, 4, p)$ by lifting S_2.
(4) Take T_1 as the smallest set among S_1, S_2 and S_4.
(5) Compute T_3 by lifting T_1 with $c = 3$.
(6) Compute T_9 by lifting T_3 with $c = 3$.

In the second, the $c = 2$ lifts are unnessary as both programs are capable of finding the proof for each prime. However, by doing the extra steps, the amount of work left at $c = 3$ lift was reduced, shortening the compute time. By running the two methods on a 10-core Apple M4 processor (Macbook Air 2025), the first method takes 18 seconds while the second methods uses 12 seconds.

This speedup can be dramatic in certain cases, for example in the case of $k = 9$. One idea is to generate $S_1 := I'(9, 1, p)$, then lift it to $S_2 := I'(9, 2, p)$ with $c = 2$ and lift it to $S_{10} := I'(9, 10, p)$ with $c = 5$. My proposal is

(1) Generate $S_1 := I'(9, 1, p)$
(2) Compute $S_2 := I'(9, 2, p)$ by lifting S_1 with $c = 2$.
(3) Compute $S_4 := I'(9, 4, p)$ by lifting S_2 with $c = 2$.
(4) Take T_1 to be the smaller set among S_2 and S_4.
(5) Compute T_3 by lifting T_1 with $c = 3$. At this point, T_3 can be either $I'(9, 6, p)$ or $I'(9, 12, p)$.
(6) Take U_1 to be the smaller set among T_1 and T_3.
(7) Compute U_5 by lifting U_1 with $c = 5$. At this point, $U_5 = I'(9, l, p)$ for some l such that $10 \mid l$.

For a specific number, the first method without the intermediate step could not compute $I'(9, 10, 53)$ within 15 minutes while it takes under 5 seconds with the second method. This opens up the possibility of lifting an improper set to a larger prime number and could help with the verification of the conjecture for 11 runners.

As discussed in Section 7 of [7], a proof of the conjecture in 11 runners case cannot avoid a lift of $I(k, l, p)$ to $I(k, 11l, p)$, which requires verification of 11^{10} candidates per element in $I(k, l, p)$. Using the seive steps and focusing at $I'(k, l, p)$ instead of $I(k, l, p)$ can reduces the number of elements at interest, resulting in less computation. Another technique we can employ is projection. For example once we obtain $I'(k, l, p)$, projecting the speed sets back into modulo p using the natural $\pi : \mathbb{Z}/pl\mathbb{Z} \to \mathbb{Z}/p\mathbb{Z}$ collapses some the improper sets from $I'(k, l, p)$, maintaining only the structure in $\mathbb{Z}/p\mathbb{Z}$. This action loses fined grain detail of each improper speed set but reduces the number improper speed set that need calculation.

The projection step is legal because of the following lemma 5

Lemma 5. *For a fixed k and p, let $\pi_{a\to b} : (\mathbb{Z}/ap\mathbb{Z})^k \to (\mathbb{Z}/bp\mathbb{Z})^k$ be the coordinate-wise extension of the natural projection whenever $b \mid a$, then,*

$$I'(k, nl, p) \subseteq \pi^{-1}_{nl\to l}\big(\pi^{-1}_{l\to 1}(\pi_{n\to 1}(I'(k, n, p))) \cap I'(k, l, p)\big)$$

Proof. By Lemma 3.3 of [7], the projection maps satisfy

$$I'(k, nl, p) \subseteq \pi^{-1}_{nl\to n}(I'(k, n, p))\,, \tag{1}$$

$$I'(k, nl, p) \subseteq \pi^{-1}_{nl\to l}(I'(k, l, p))\,. \tag{2}$$

Let $v \in I'(k, nl, p)$ be arbitrary. From (1), we have $\pi_{nl\to n}(v) \in I'(k, n, p)$, hence

$$\pi_{n\to 1}(\pi_{nl\to n}(v)) \;\in\; \pi_{n\to 1}(I'(k, n, p))\,.$$

Since $\pi_{l\to 1} \circ \pi_{nl\to l} = \pi_{nl\to 1} = \pi_{n\to 1} \circ \pi_{nl\to n}$,

$$\pi_{l\to 1}(\pi_{nl\to l}(v)) \;\in\; \pi_{n\to 1}(I'(k, n, p))\,,$$

i.e. $\pi_{nl\to l}(v) \in \pi^{-1}_{l\to 1}(\pi_{n\to 1}(I'(k, n, p)))$. Combined with (2), which gives $\pi_{nl\to l}(v) \in I'(k, l, p)$, we obtain

$$\pi_{nl\to l}(v) \;\in\; \pi^{-1}_{l\to 1}\big(\pi_{n\to 1}(I'(k, n, p))\big) \;\cap\; I'(k, l, p),$$

which completes the proof.

□

Intuitively, the lemma ensures that any improper speed set at level nl must also be present at level l, so projecting down $I'(k,l,p)$ to $I'(k,1,p)$ cannot discard anything that genuinely belongs in $I'(k,nl,p)$.

Result

Rosenfeld wrote [3] to verify the conjecture for eight runners and Trakulthongchai developed [8] to verify the conjecture of nine and ten runners. His computation takes 15 minutes for $k=8$ and approximately 23 hours for $k=9$. Building on the framework of [8], I developed [5], with assistance from AI-based tools, to improve performance and scalability.

Using the improvements proposed, I managed to complete the proof of $k=8$ in 12 seconds and $k=9$ in 181 seconds on a 10-core Apple M4 processor (Macbook Air 2025). The proof of $k=10$ and $k=11$ were run in multiple batches, multiple versions, and multiple machines, thus making the exact time taken incomarable with other cases. The source of the computation, along with log files, can be found in [5]

For the eleven runners case, let

$$P_{10} := \left\{ \begin{array}{r} 103, 131, 137, 139, 149, 151, 157, 107, 109, 127, 163, 167, 173, \\ 179, 433, 191, 193, 197, 199, 211, 223, 227, 229, 233, 239, 241, \\ 251, 257, 263, 269, 271, 277, 281, 283, 293, 307, 311, 313, 317, \\ 331, 337, 347, 349, 353, 359, 367, 373, 379, 383, 389, 401, 409, \\ 419, 421, 431, 433, 439, 443, 449, 457, 461 \end{array} \right\}$$

and for the twelve runners case, let

$$P_{11} := \left\{ \begin{array}{r} 23, 131, 137, 139, 149, 151, 157, 163, 167, 173, 179, 181, 191, \\ 193, 197, 199, 211, 223, 227, 229, 233, 239, 241, 251, 257, 263, \\ 269, 271, 277, 281, 283, 293, 307, 311, 313, 317, 331, 337, 347, \\ 349, 353, 359, 367, 373, 379, 383, 389, 397, 401, 409, 419, 421, \\ 431, 433, 439, 443, 449, 457, 461, 463, 467, 479, 487, 491, 499, \\ 503, 509, 521, 523, 541, 547, 557, 563, 569, 571, 577 \end{array} \right\}$$

Using the algorithm described in the previous section, we can verify that for every prime $p \in P_{10}$ there exists l such that the set $I(10,l,p)$ is empty. Equivalently, any counterexample to the 11 runners case of the Lonely Runner Conjecture must have speeds divisible by each prime in P_{10}. Consequently, if such a counterexample exists, its product of speeds must be divisible by the product $\prod_{p \in P_{10}} p$

Similarly, we can also verify that for every $p \in P_{11}$ there exists l that makes $I(11,l,p)$ empty. Therefore, if a counterexample of the Lonely Runner Conjecture exists for the $k=11$ case, the product of speeds must be divisible by $\prod_{p \in P_{11}} p$.

Theorem 6. *The Lonely Runner Conjecture holds for $k=10$ and $k=11$*

Proof. As Trakulthongchai has shown in [7] that the Lonely Runner Conjecture holds for $k=9$, it follows from the result of Rosenfeld [4] that it suffices to check that the conjecture holds for integral speeds up to B_{10} to conclude that the conjecture holds for $k=10$. From computation

$$B_{10} < e^{337.65} < e^{339} < \prod_{p \in P_{10}} p$$

it can be concluded that the Lonely Runner Conjecture holds for $k=10$.

Since it holds for $k = 10$, it suffices to check if the conjecture holds for integral speeds up to B_{11}, and we achieved

$$B_{11} < e^{434.5} < e^{435} < \prod_{p \in P_{11}} p$$

form the described algorithm. Hence, the Lonely Runner Conjecture holds for $k = 11$. □

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